

MNLP Report Homework 1

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1 Introduction

The NO-LLM part is conducted via multimodal approach combining embedding textual data of name + description and graph analysis for items.

The LLM part is based on the addition of a new feature and prompt engineering.

2 Methodology

2.1 NO-LLM

After the inspection of the dataset, we first decided using the richest textual features (name and description) into a single text input for embeddings. To generate them, we used the text-embedding-3-small model from OpenAI that produces 1536 length vector embeddings. Then we focused on the items that are used to build a graph, composed as follows: for each QId, a sparql query fetches up to 10 property-value pairs, representing edge and child respectively.

The idea behind this is to exploit the relationships between entities and their instances. Two metric similarities, representing how structurally similar each item is to others, are computed as the mean and maximum neighbor similarity. Intuitively, if two nodes are entities having similar instances, thus similar neighbors, then the scores will be higher. An example of a node in the graph is shown in Fig. 1.

Then both approaches are combined to perform the classification task. Textual embeddings and similarity score concatenations build training and validation sets.

Through TSNE reduction we observed how classes were distributed along the feature space, visible in Fig. 2. The Cultural Agnostic (blue) class is well-clustered, while Cultural Exclusive (orange) and Representative (green) are not. Infact, the items that belong to these classes often blur the lines between cultures, and the distinction between them may not be clear-cut.

To find out the best approach for classification, multiple models such as KNN, Logistic Regression, SVM and XGB were evaluated and after comparison, the KNN algorithm was selected due to its superior performance. K is finetuned and set to 20.

2.2 LLM

In this section, we introduced an additional feature indicating the original language of each Wikidata item. This information was retrieved using a SPARQL query to the Wikidata endpoint. If the original language could not be determined, English was assigned as the default.

Then, we decided manipulating the input of the LLM by applying prompt engineering. We used 6 tags [LANG], [NAME], [DESC], [TYPE], [CAT], [SUBCAT] as input information for the LLM. For each item of the dataset the prompt generated was in the form "You are a classification expert. Based on the information below, classify the item into the correct category: [LANG] ... [NAME] ... [DESC] ... [TYPE] ...".

Furthermore, [LANG], [NAME], [DESC], [TYPE] tags are used as special tokens to the xlm-roberta-base tokenizer.

The model used for classification is thus xlm-roberta-base with R-Drop Loss that is a combination of cross entropy and symmetric KL with clamped probabilities trying to reduce variance and improve generalization.

Considering overfitting as a possible drawback of LLMs, we employed checkpoints to save the best model got so far.

All parameters set during training are reported in Table 2.

3 Experiments

3.1 NO-LLM

Before using the methodology described in Sec. 2.1, we tried different approaches regarding embeddings and models.

Particularly we used different combinations of features to build the embedding and distinct embedding techniques such as Word2Vec and TF-IDF. We also tested several models to perform classification: KNN, Logistic Regression, SVM and XGB.

3.2 LLM

Before using the methodology described in Sec. 2.2, we performed several trials.

First, we augmented the dataset by generating new instances for underrepresented categories.

We also tried adding a new field text in the dataset with a more precise and long description of each item. Both the above techniques caused overfitting. Then, we tried different approaches of prompt engineering. Before applying this technique, we computed the mutual info classif method, measuring the amount of information that knowing the feature provides about the target variable.

Based on this, we combined the most important features into the prompt input of the LLM. We also tested them with different models:

- Bert
- RoBerta-base
- RoBerta-large
- DistilBert
- Albert

4 Results

4.1 NO-LLM

The results on the dev set are reported in the Table. 1. The model is good at identifying Cultural Agnostic and has a balanced trade-off between precision and recall, but struggles more for Cultural Representative and Exclusive. Those results are not surprising since the evidences discussed at Section 3.1.

4.2 LLM

The results on the development set are presented in Table 3. As expected, the model demonstrates improved performance across all three classes, achiev-

ing a more balanced trade-off among the evaluation metrics.

References

A Appendix

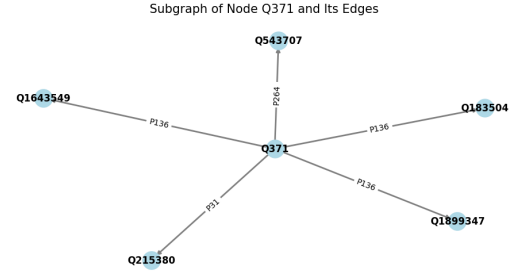


Figure 1: Graph Node Example

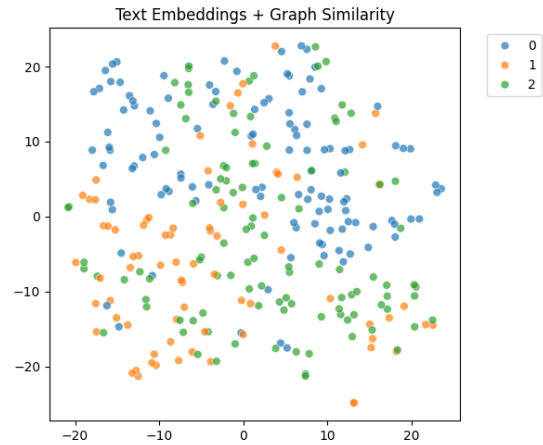


Figure 2: Text Embedding+Similarity Score

Class	Precision	Recall	F1-score
CA	0.89	0.93	0.91
CE	0.59	0.75	0.66
CR	0.76	0.57	0.65
Acc			0.76
Macro	0.75	0.75	0.74
Weighted	0.77	0.76	0.75

Table 1: Classification report (NO-LLM)

Hyperparameter	Value
Learning Rate (lr)	2×10^{-5}
Weight Decay	0.01
Batch Size	16
Epochs	10
Optimizer	Adam
Gradient Clipping	0.9

Table 2: Training hyperparameters

Class	Precision	Recall	F1-score
CA	0.89	0.95	0.92
CE	0.75	0.62	0.68
CR	0.72	0.76	0.74
Acc	-	-	0.80
Macro	0.79	0.77	0.78
Weighted	0.79	0.80	0.79

Table 3: Classification report (LLM).